

Solar-Powered Water Treatment for Agriculture and Drinking Water in Rajasthan, India

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"Water Crisis in Rajasthan & Impacts on Agriculture and Public Health [1,2]"

The Challenge in Rajasthan

- Drinking water contamination
- Crop contamination & soil degradation
- Reduced yields & income loss
- Increased public health risks
- Failure to meet quality standards

Sources of Water Contamination

- Agricultural runoff (**fertilizers & pesticides**)
- Livestock waste entering open channels
- Industrial discharge
- Open, unprotected irrigation channels
- The average total dissolved salts (TDS) is in the range of 750-2500 mg/L. (very high compared to WHO standards of <600 mg/L).

Impacts on People

- Exposure to bacteria (E. coli), fluoride, arsenic
- Higher disease rates in rural communities

Impacts on Agriculture

- **Soil quality** decline & salinity
- **Crop contamination**
- Reduced yields and lower income
- Lower market value; harder to meet standards

Why Current Systems Fail [3,4,5]

No Affordable Water Treatment

Most rural areas still lack low-cost systems to clean irrigation and drinking water.

Unreliable Electricity

Water pumps and treatment units can't run consistently because the village power supply is unstable.

Groundwater Depletion

Overuse of groundwater lowers water tables and increases contamination risk.

Poor Water Quality Monitoring

Rural water sources are rarely tested, so polluted water continues to be used daily.



“Our Solution: Solar-Powered Water Treatment System”

- Use excess solar energy to run a local water-treatment unit
- Solar pump → raw water
- Multi-stage filtration for irrigation & drinking use
- Supports both agriculture & communities
- Converts surplus solar power → reliable clean water

OBJECTIVES

100% Solar-Powered

From inlet to outlet, Solar Energy will power the entire treatment system.

This includes:

- Inlet & Outlet Pumps [6]
- Flow Control Valves [6]
- Dosing Control Valves [6]
- Necessary motors [6]

Dual Stream Treatment

Surface water will be treated to two standards:

- Irrigation Water (Large Volume)
- Potable Water (Small Volume)

This ensures:

- Energy Savings [7]
- Chemical Savings [7]
- Material Savings [7]
- Drinking Water Access [8]

Long Term Impacts

Treating irrigation water at the source reduces pollutant loads entering open channels. Over time, as cleaner water continually flows through the system:

- groundwater recharges [9]
- contaminant buildup decrease [9]
- soils gradually recover [9]
- Crop contamination decreases [9]
- overall water quality in the region steadily improves [9]

Main Contaminant: *Agricultural Runoff (Ammonia)*

Trickling Filter (Fixed Film)

Specifications:

- Effective ammonia removal (70-95%) [10]
- Minimal operator training [10]

Main Contaminant: *Livestock Waste (Pathogens)*

UV Disinfection

Specifications:

- 90-99% inactivation of Viruses, Bacteria & Protozoa [11]
- Can be powered by solar [6]

Chlorine Disinfection

Specifications:

- 90-99% inactivation of Viruses & Bacteria [12]
- NOT effective for Protozoa [12]
- Calculated dosing requires training [12]

Other Considerations: *TDS*

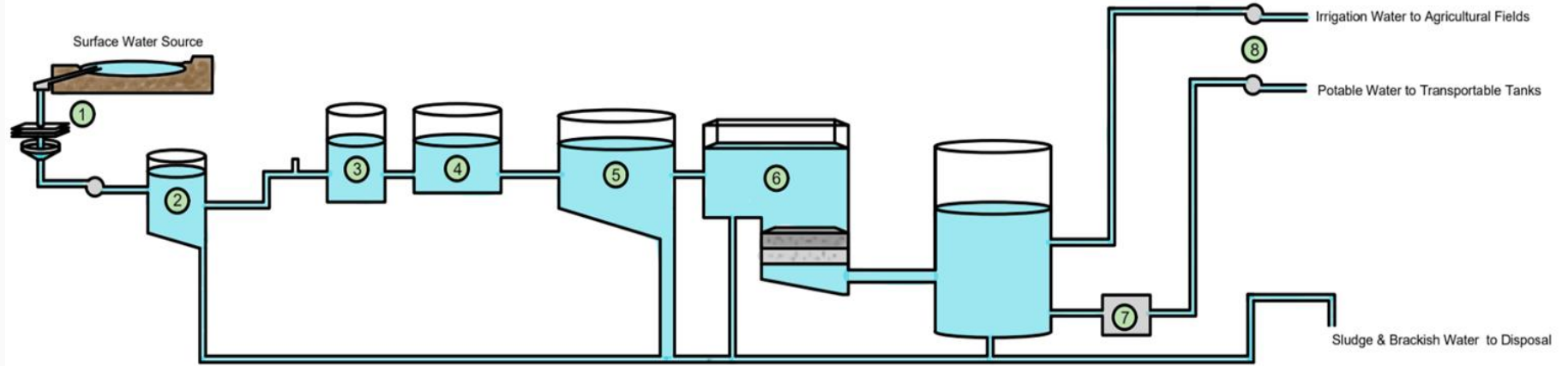
Total Dissolved Solids

- **Effect taste/aesthetics [9]**
- **Can effect soil quality [9]**

Specifications:

- **Must use high energy processes (RO, Nanofiltration, electrodialysis) [6]**
- **Requires training [6]**
- **Should only be installed based on soil quality in the area. [9]**

Water Treatment Design



1 Intake & Screening

2 Pre-Sedimentation

3 Coagulation

4 Flocculation

5 Sedimentation

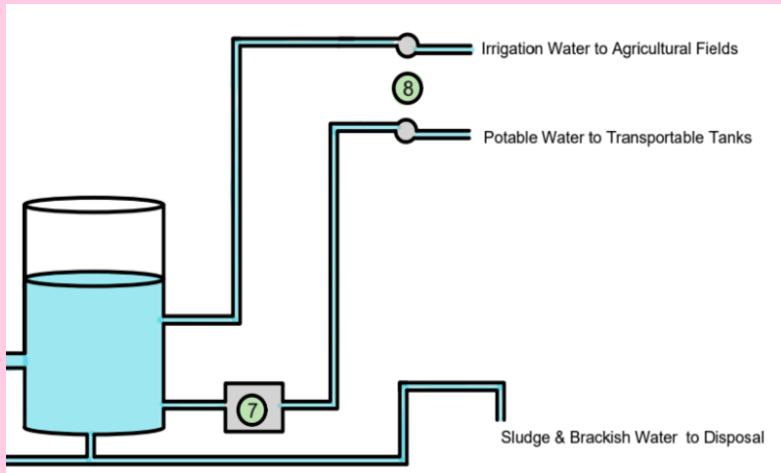
6 Filtration

7 Disinfection (Potable Only)

8 Distribution

The Split:

Occurs after filtration, and before disinfection



Irrigation Stream

- Does not require disinfection [8]
- Pumped to existing irrigation systems [13]
- Slowly Decontaminates Crops [9]
 - Soil Remediation
 - Groundwater recharge and remediation

Potable Stream

- Meets drinking water requirements [12]
- Pumped to transportable tanks [13]
- Ensures safe drinking water for rural communities [8]

Plant Design - Site Specifications

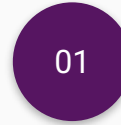
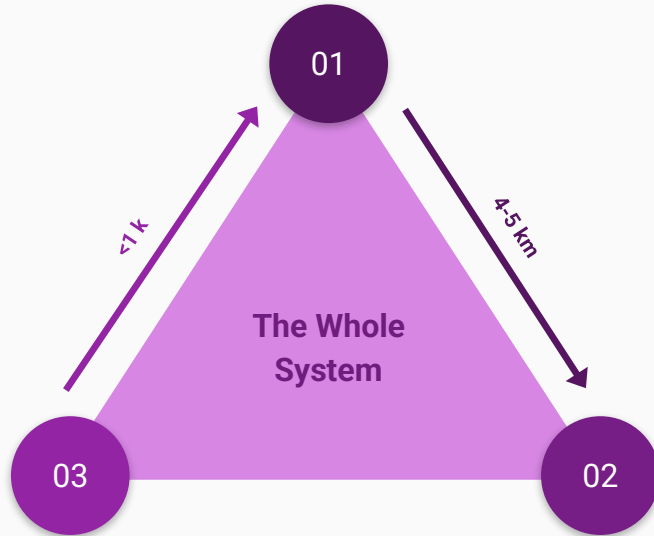
Energy

- Situated near one of the world's largest solar park named Bhadla Solar Park, Rajasthan, India. [16]
- Typical GHI of $\sim 5\text{-}7 \text{ kWh/m}^2/\text{day}$.
- "Full sun"/ peak hours of $\sim 5.5\text{-}6 \text{ h}$.
- Generation capacity of 2245 MW.
- Generates 25-40% surplus energy than the current energy demand, leading to curtailment.

Water

- Situated near the banks of Biggest Canal Project named Indira Gandhi Canal. [17]
- Discharge capacity of $460 \text{ m}^3/\text{sec}$.
- Canal consists of nearly 650 km long main feeder canal alongside 9000 km long distribution channels
- Serving approx. 1.4 Million hectares of cultivable land across the north-western of rajasthan.

Source to Distribution



On-Site Water Treatment Plant



Distribution

- On-Site storage for drinking water, to be distributed as needed via truck (contingency)
- Community over-head tanks and pipings.
- Central pipeline/ existing distribution canals for irrigation water.



Source Water

- Indira Gandhi Canal

Plant Design Considerations - Water

Drinking Water

- For a community of 5000 people.
- As per “Jal Jeevan Mission”, An Indian Government scheme guideline of 55 LCPD(liters per capita per day) [18]
- $5000 \times 55 \text{ LCPD} = 275,000 \text{ LPD} \sim \mathbf{275 \text{ m}^3/\text{day}}$.

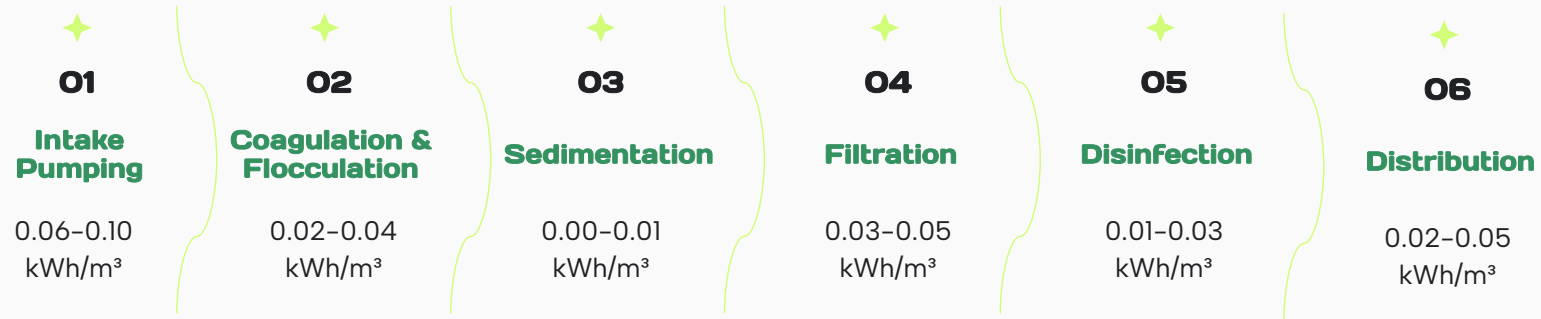
Irrigation

- For 100 hectares of cultivable land.
- Approximate requirement of **30 $\text{m}^3/\text{hectare}/\text{day}$** [19](*drought tolerant crops - 20-35 $\text{m}^3/\text{hectare}$; common cereals and pulses - 25-40 $\text{m}^3/\text{hectare}$; water intensive crops - 35-55 $\text{m}^3/\text{hectare}$)*
- $100 \text{ hectares} \times 40 \text{ m}^3/\text{hectare}/\text{day} = \mathbf{3000 \text{ m}^3/\text{day}}$.

Total Water Consumption = $3275 \text{ m}^3/\text{day}$

Required Energy

6 Components-wise Energy Consumption [7]:



Cumulative Energy Requirement: 0.28 kWh/m³ (~0.3 kWh/m³).

Total Energy Requirement= $3275 \times 0.3 = 982.5$ kWh/ day (~1 MWh/day)

Cost and Economic Value Analysis of the Proposed Solution

All economic calculations were conducted assuming a 25-year lifespan, a 7% discount rate, and a 2% price escalation rate.

KEY ASSUMPTIONS

1

Project Lifespan

A project lifespan of 25-years is assumed for this economic analysis.

2

Discount and Price Escalation Rates

Discount Rate: 7%
Price Escalation: 2%

3

Cost of Land

Cost of land is neglected due to assumptions of government cooperation.

4

Distance Traveled

Averages are assumed for truck distances, due to unclear delivery routes.

5

Revenue Model

This project is not intended to be profitable, and will operate on a government funding model. Value added will be discussed in terms of societal benefit.

UNIT OPERATIONS

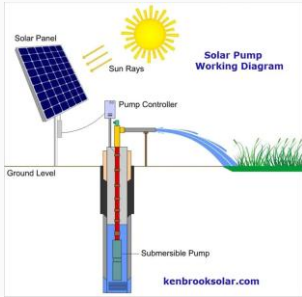
1

Water Pump [20]

System of water pumps to retrieve ground or surface water from selected site.

- Used local solar-powered pump as model, although we will draw power from grid overproduction

- Unit Cost:** \$4,756
Total Number of Pumps: 33
Unit Pump Capacity: 100 m³/day



2

Treatment Process [21]

Overall water treatment plant.

- Used a metric to estimate cost based on capacity of plant.

- Total Installation Cost:** \$13 million



3

Water Storage Tank [22]

To hold 2 days of capacity for drinking water only, a 600 m³ tank is installed.

- Used a scaled-up model of a real-life tank to estimate cost of our tank.

- Total Installation Cost:** \$103,250



4

Tanker Trucks [23]

Large tanker trucks will deliver the water directly to the customer.

- Used the BharatBenz 4228R Tanker as our model.

- Unit Cost:** \$44,400
Total Number of Trucks: 2
Unit Tanker Capacity: 280 m³



5

Irrigation Pipeline [24]

We assume a 5 km pipeline will be needed to pump irrigation water to the farms.

Estimated total cost is \$83,000 which is a scaled amount based on similar-scale projects in India.



Capital Expenditures (CapEx)

\$13 million	Water Treatment Plant	Mixed process water treatment plant.
\$103,250	Storage Tank	600 m ³ capacity permanent storage tank.
\$88,800	Tanker Trucks	2 trucks.
\$0.45 million	Water Pumps	33 pumps, each replaced 4 times over 25 years.
\$2.74 million	Contingency Cost	Accounting for changing prices and other expenses. Assumed to be 20% of base CapEx.
\$16.47 million	TOTAL CAPEX	Total initial capital investment

Annual Operating Expenditures (OpEx)

\$1.3 million	Water Treatment Plant	Estimated as 10% of CapEx.
\$10,325	Storage Tank	Estimated as 10% of CapEx.
\$528.82	Tanker Trucks	Cost of gas, assuming 3.12 mi/day of travel for each truck.
\$1.70 million	Water Pumps	Estimated as 38% of total life cycle cost.
\$376,407.38	Contingency Cost	Accounting for changing prices and other expenses. Assumed to be 12.5% of base CapEx.
\$3,387,666.40	TOTAL OPEX	Total yearly operating expenditures.

Lifetime Cost:

\$55,943,805.39

This cost represents the sum of NPV-adjusted operating expenditures and capital expenditures over the 25 year project lifespan.

Energy Cost Saved

\$0.084/kWh

Cost of Energy [25]

\$753,086.25

Lifetime Energy Cost

For the small sample case. However, this plant could handle a much larger area, which means the current capacity underestimates the service area.

Carbon Emissions [26]

2,659.82 kg
CO₂/year

Emissions per year

66,495.45 kg
CO₂

Lifetime emissions

The only source for carbon emissions is from the truck used for potable water distribution, which could be reduced or eliminated by using pipes instead of trucks.

Potential Improvements

Water Transport

- The use of trucks for water transport, while important for reaching more rural areas, reduces the overall sustainability of the solution.
- Future iterations should explore feasibility of piping water to locations, or having a centrally located “water store”.

Potable Water

- In regards to profitability, irrigation water provided essentially zero revenue to the project.
- Future iterations could explore how to increase the amount of potable water produced to improve profitability.

Water Quality Improvement Timeline

0-1 Seasons	Immediate Impact	• Farmers start irrigating with clean water. • New contamination of crops from irrigation water is sharply reduced. • Canal water quality improves as clean water displaces polluted water.
2-3 Seasons	Short-to-Medium Term	• Repeated leaching with cleaner water reduces the concentration of many salts and mobile pollutants in the root zone. • Crop quality and yields start to reflect a cleaner baseline environment, not just cleaner irrigation events.
5-10 Years	Long Term	• The area's overall water quality (canals, soils, shallow groundwater) shifts to a new, cleaner equilibrium. • This long-term improvement will meet stricter export standards and create a reputation for high-quality, low-contaminant produce.

Countries with strict import regulations often reject low quality crops, with standards Rajasthan struggles to meet.

This includes things like microbial contamination, pesticide or heavy-metal traces, excessive sediment or spoilage risk, and signs of polluted irrigation sources.

ECONOMIC IMPACTS

1

Increase in Crop Yields

By introducing cleaner irrigation water that meets these requirements, previous studies indicate that yields could be increased by 13-41% [27].

2

Farmer Income

Improving efficiency and quality of irrigation could increase farmer profitability by 53-115% [27][28].

3

Water Use Efficiency

Improvements in irrigation technology and water quality could see 39-55% improvements in water use efficiency [27].

4

Foreign Investment

The improvement in water quality may lead to an increase in foreign and domestic investment in Rajasthan.

5

Cost of Inaction

Failure to improve water productivity is estimated to have a potential cost of upwards of \$2.5 billion for the government of India by 2050 [29].

SELF-REINFORCING ECONOMIC CYCLE

Excess Solar Energy

By utilizing excess solar energy, this process introduces no new burden on the electrical grid and operates entirely on previously wasted energy.

The excess solar energy then powers the water treatment process, which feeds into the overall economic loop.

Clean Water

The treatment of polluted ground and surface water has two major benefits:

- Improved crop quality
- Improved access to drinking water

In a region with high levels of groundwater contamination, high-quality water is extremely valuable.

Additionally, improved access to clean drinking water contributes to regional public and environmental health.

High-Value Agriculture

The cycle continues through increasing the value of crops in the region:

- Higher crop standards unlock export markets.
- Increased revenue drives more investment in solar and water systems.

As investment in solar and water systems increases, the value and applicability of this system increases, meaning the system creates its own value.

This self-sustaining economic loop creates long term resilience.

This solution is, at a fundamental level, economic, agricultural, environmental, and community focused.

Thank you!

Questions?

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